Ethiopia B2B E-Commerce Platform

Enterprise Wholesale Distribution Ecosystem Engineered for Ethiopia

Project Scope & Technical Foundation

- Technology Stack: Medusa v2 Headless Engine, Next.js 14 SSR, React Native Expo
- Target Market: Addis Ababa Central Warehouse & Nationwide Ethiopian Distribution
- Core Currency: Ethiopian Birr (ETB / Br) - zero foreign exchange conversion friction
- B2B Governance: Corporate customer accounts, buyer limits, and approval workflows
- Omnichannel Delivery: Responsive web procurement portal + iOS & Android mobile app

Presenter: Software Engineering Intern | Academic & Supervisor Evaluation

Executive Summary & Project Vision

Bridging the Digital Divide for Wholesale Commerce Across Ethiopia

Core Pillars of the Platform

* 100% Ethiopian Localization: ETB currency, +251 phone, Addis Ababa warehouse, and domestic shipping routes.

* Multi-Seat Corporate Structure: Business onboarding, employee permissions, and spending limit enforcement.

* Request For Quote (RFQ): Digital volume negotiation with automated draft order generation.

* Tiered Commercial Pricing: Bronze, Silver, Gold, and Platinum discount brackets rewarding loyal buyers.

* Omnichannel Architecture: Web and native mobile apps.

Zero Currency Friction

All listings, order lines, and invoices are strictly in ETB. Eliminated USD foreign exchange and banking conversion errors.

Enterprise Governance

Budget controls, spending caps, and dual-layer approvals ensure zero unapproved expenditure for enterprise customers.

Cloud Deployment

Live Medusa backend on Render, PostgreSQL database, Next.js web portal on Vercel, and cross-platform Expo mobile app.

Wholesale Commerce Challenges in Ethiopia

Why Generic Global E-Commerce Platforms Fail in the Ethiopian Market

1. Foreign Currency & FX Overhead

- Standard software assumes USD or EUR with forced forex.
- Severe Ethiopian FX regulations make USD pricing illegal and completely impractical for domestic B2B trade.
- Causes massive accounting reconciliations and failures.

2. Absence of Corporate Governance

- B2C carts assume individual retail credit cards.
- No concept of employee spending limits or branch budgets.
- Company owners cannot delegate purchasing authority without risking unauthorized corporate expenses.

3. Friction in Bulk Price Negotiation

- Wholesale orders require volume discounting (RFQ).
- Offline negotiation via phone calls or WhatsApp takes days.
- Leads to miscommunication, lost quotes, and inventory conflicts during long manual negotiation cycles.

4. Desktop vs Field Mobile Disconnect

- Ethiopian merchants (Merkato, Bole) operate on smartphones.
- Traditional desktop web stores fail to engage mobile buyers.
- Need native cross-platform mobile apps with fast catalog search, offline resilience, and instant quote submission.

End-to-End System Architecture

Decoupled Headless Micro-Modular Stack for Maximum Reliability

Tier 1: Client & Presentation Layer

- Next.js 14 SSR Web Storefront: High-speed server rendering, corporate login, checkout.
- Addis B2B Mobile App (React Native Expo): Native iOS & Android, bulk steppers, mobile RFQ.

Tier 2: Business Logic & Headless Engine (Medusa v2)

- Custom Company Module: Hierarchical company profiles, spending rules, employee limits.
- Approval Engine: Cart interception workflows, email alerts, multi-stage approval states.

Tier 3: Database & Cache Infrastructure

- PostgreSQL 16: Relational schema storing companies, employees, orders, and quotes.
- Redis Pub/Sub: Real-time event broadcasting and high-performance session caching.

Tier 4: Ethiopian Localization & Cloud Deployment

- Deployed on Render Cloud with automated CI/CD and secure PostgreSQL environment.
- Sole currency: ETB (Birr), Addis Ababa central warehouse, Ethiopian domestic courier rates.

Distributor-to-Business Hierarchy Model

Multi-Tenant Organizational Structure with Fine-Grained Spending Governance

Demo Business Customers in Ethiopia

1. Bole Electronics Retail (Addis Ababa)
   - Tier: Gold (10% Discount)
   - Spending Limit: 500,000 ETB / month

2. Merkato Tech Solutions (Addis Ababa)
   - Tier: Silver (5% Discount)
   - Spending Limit: 300,000 ETB / week

3. Hawassa Industrial Corp (Hawassa)
   - Tier: Platinum VIP (15% Discount)
   - Spending Limit: 1,000,000 ETB / month

4. Bahir Dar Office Supplies (Bahir Dar)
   - Tier: Bronze (Standard Pricing)
   - Spending Limit: 150,000 ETB / month

** Seed script automatically registers all demo companies.*

Employee Role Permissions & Limits

Company Admin (e.g., Alem Tesfaye):

- Full organizational visibility and employee onboarding.
- Spending Limit: 500,000 ETB/month.
- Authorizes or denies pending purchase requests.

Purchasing Officer / Buyer (e.g., Kebede Alemu):

- Browses verified wholesale catalog with Birr pricing.
- Spending Limit: 100,000 ETB/month.
- Can place direct orders within limit; triggers approval above.

Flexible Limit Windows

The system supports 4 flexible reset schedules:

Daily, Weekly, Monthly, or Yearly spending caps.

Multi-Stage Approval Workflows

Automated Spending Controls Eliminating Unauthorized Corporate Expenses

Step 1: Order Creation & Limit Verification

- Employee Kebede Alemu adds 10x Laptops to cart (Total: 1,299,000 ETB).
- System evaluates active spending rule: 1,299,000 ETB exceeds employee limit of 100,000 ETB.
- STATUS: Automatically transitions cart to "REQUIRES_APPROVAL" without debiting customer.

Step 2: Company Admin Approval Gate

- Company Admin (Alem Tesfaye) receives real-time notification with order summary and justification.
- Admin reviews department budget allocation and necessity in the dashboard.
- One-click decision: Approve (advances order) OR Reject (cancels order with explanatory note).

Step 3: Distributor Commercial Confirmation & Warehouse Pick

- For high-volume enterprise orders, distributor sales team performs stock allocation check.
- Addis Ababa warehouse generates domestic pick-list and dispatches ground freight.
- Complete audit trail recorded: timestamps, approver identity, and invoice in ETB.

B2B Quote Negotiation Engine (RFQ)

Empowering High-Volume Price Negotiation for Enterprise Hardware Deals

RFQ Negotiation Lifecycle

1. Request Submission:
   Corporate buyer requests custom rate for 100x monitors (Catalog list price: 5,990,000 ETB).

2. Distributor Review:
   Distributor evaluates stock in Addis Ababa warehouse and issues counter-offer of 5,400,000 ETB (590k savings).

3. Acceptance & Order Conversion:
   Buyer accepts counter-offer via portal or mobile app; system locks quoted price and generates draft order.

4. Fulfillment:
   Invoice generated in ETB; goods dispatched to client.

Engineering Advantages of Custom RFQ

- Inventory Reservation Lock:
  Prevents overselling by placing soft lock on warehouse stock during active negotiation window.

- Full Audit Trail & Messaging:
  Preserves complete timestamped dialogue between distributor and client for procurement compliance.

- Mobile Native Experience:
  Buyers can submit, inspect, and approve quote offers directly from their smartphones via Addis B2B Mobile App.

Tiered Customer Pricing & Catalog in ETB

Rewarding Enterprise Loyalty & Volume Purchases with Transparent Birr Pricing

Bronze Tier

Standard Base Price

Silver Tier

5% Commercial Discount

Gold Tier

10% Commercial Discount

Platinum VIP

15% Commercial Discount

Sample Hardware Catalog (ETB)

- 16" Pro Laptop: 129,900 ETB (Gold: 116,910 ETB)
- Smartphone (256GB): 99,900 ETB (Gold: 89,910 ETB)
- 34" Curved Monitor: 59,900 ETB (Gold: 53,910 ETB)
- Enterprise Headset: 14,900 ETB (Gold: 13,410 ETB)
- HD Conference Webcam: 3,200 ETB (Gold: 2,880 ETB)
- Wireless Desktop Keyboard: 9,900 ETB (Gold: 8,910 ETB)

Automated Bulk Quantity Breaks

Stacked on top of customer tiers:

- 1 - 10 Units: Standard customer tier pricing.
- 11 - 50 Units: Additional 3% volume rebate.
- 51 - 100 Units: Additional 5% volume rebate.
- 100+ Units: Additional 8% rebate or instant RFQ trigger.

** Currency Symbol Map prioritized "Br" for ETB across the app.*

Addis B2B Mobile App (React Native + Expo)

Delivering Enterprise Wholesale Commerce to Mobile Devices Across Ethiopia

Live Cloud Backend

- Directly connects to live Render Medusa v2 backend via publishable store keys.
- Real-time stock queries from Addis Ababa warehouse.
- Dynamic Birr prices with customer group tiers.

Mobile Wholesale UX

- Bulk quantity steppers for wholesale batches (10x, 50x, 100x).
- Multi-variant selector: RAM, storage, colors.
- Dedicated Request Quote modal for field agents.

Cross-Platform Expo

- Single TypeScript codebase running on iOS & Android.
- Instant testing via QR code in free Expo Go app.
- Corporate employee login displaying budget limits.

Key Contributions & Engineering Tasks

Hands-on Software Development, Architectural Customization, and Documentation

Backend & Data Engineering

- Currency Map Customization:
  Registered 'etb: "Br"' in currency-symbol-map.ts as priority.

- Database Seed Pipeline:
  Refactored seed.ts to purge EUR/USD and establish ETB.

- Geographic & Warehouse Setup:
  Configured Ethiopia-only region and Addis Ababa warehouse.

- Demo Company Automation:
  Built automated script seeding 4 Ethiopian businesses & staff.

Storefront, Mobile & Documentation

- Addis B2B Mobile App:
  Engineered React Native mobile client connecting to Render API.

- Storefront Localization:
  Formatted Next.js checkout for Ethiopian phone & addresses.

- Cloud Deployment Configurations:
  Documented Vercel and Render deployment procedures.

- 1,500+ Lines of Technical Documentation:
  Authored ETHIOPIA_SETUP.md, DEMO_CHECKLIST.md, and visual models.

Technical Challenges & Solutions

Overcoming Architectural, Modeling, and Integration Obstacles

Challenge 1: Medusa v2 Micro-Kernel Remote Link Architecture

- Problem: Joining Company entities to Customer Accounts without altering core tables.
- Solution: Utilized Medusa's Remote Link engine to establish loosely coupled virtual joins between company_id and customer_id, maintaining architectural modularity.

Challenge 2: Approval State Transitions & Cart Reservation

- Problem: Orders exceeding employee budgets risk dropping cart state or triggering premature charges.
- Solution: Created an intermediate 'REQUIRES_APPROVAL' status holding items in soft reserve without executing payment gateway capture until corporate admin signs off.

Challenge 3: Multi-Currency Artifacts in Open-Source Core

- Problem: Open-source commerce templates default to hardcoded USD/EUR price matrices.
- Solution: Re-engineered seed scripts, purged non-ETB prices, and established strict single-currency validation rules preventing foreign currency insertion.

Demonstration Scenarios & Validation

Empirical Testing Across Routine, Budget-Exceeded, and Wholesale RFQ Use Cases

1. Standard Checkout

Actor: Buyer Kebede
Company: Bole Electronics

Order: 2x HD Webcams
Total: 6,400 ETB
Limit: 100,000 ETB

Result: PASSED

- Instant checkout success.
- Warehouse notified.

2. Spending Limit Halt

Actor: Buyer Kebede
Company: Bole Electronics

Order: 10x Pro Laptops
Total: 1,299,000 ETB
Limit: 100,000 ETB

Result: PASSED

- Checkouts halts safely.
- Routed to Admin Alem.
- Admin approves in panel.

3. High-Volume RFQ

Actor: Hawassa Industrial
Company: Enterprise Client

Order: 100x Monitors
List Price: 5.99M ETB
Action: Request B2B Quote

Result: PASSED

- Counter-offer: 5.4M ETB.
- Accepted on Mobile App.
- Locked draft order made.

Future Roadmap & Strategic Enhancements

Scaling the Platform to Power Regional Wholesale Commerce Across East Africa

1. FinTech Integrations

- Telebirr SuperApp SDK:
  Instant B2B settlement
  for Ethiopian enterprises.

- CBE Birr & CBE Direct:
  Commercial Bank of Ethiopia
  direct corporate transfers.

- Chapa Payment Gateway:
  Seamless multi-bank checkout.

2. Offline Mobile Sync

- SQLite On-Device DB:
  Cache catalog and client
  profiles locally on phones.

- Offline Order Queuing:
  Allow sales agents to write
  orders in low-connectivity areas.

- Auto Sync Engine:
  Background synchronization.

3. ERP & Tax Systems

- ERP Connectors:
  Bi-directional sync with
  Odoo and SAP databases.

- MOR Electronic Invoicing:
  Ministry of Revenues compliant
  electronic tax stamps.

- Automated Inventory:
  Real-time stock depletion.

Conclusion & Key Learnings

Reflections on Technical Growth, Project Impact, and Final Defense

Internship Competencies Gained

1. Headless Commerce Architecture:
   Mastered micro-modular headless systems using Medusa v2, Next.js 14 SSR, and React Native Expo.

2. Enterprise B2B Domain Engineering:
   Architected corporate hierarchies, spending limits, and automated approval workflows.

3. Software Localization at Scale:
   Customized open-source architectures to enforce strict ETB monetary policies and Ethiopian geography.

4. DevOps & Cloud Delivery:
   Deployed multi-tier backend and frontend services on Render.

Thank You!

Questions & Discussion

Ready to demonstrate:

- Live Medusa Admin Dashboard
- Company & Employee Limit Management
- Order Approval Pipeline in action
- Request For Quote (RFQ) negotiation flow
- Addis B2B Mobile App live on Expo Go